

Problem Set 4

ECON 2010 – Macroeconomics

Spring 2019

Due at the start of class Wednesday, April 3. You have **one week** to complete this problem set.

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct (some of) the problem set together in class.

Name:

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 — How to get 4% growth in the US (24 points)

President Trump has frequently discussed his desire to increase economic growth in the US. In particular, he's extolled reaching 4% real GDP growth. As he said in one 2017 speech: "They're saying three, but I say four." He's even aimed higher at times: "And I say there's no reason we shouldn't be able to get, at some point into the future, to five and above." But most often he talks about 4% as the magic number.

In 2017, real GDP growth in the US was 2.5%. You might not think 4 is that much larger than 2.5 — some countries like China have grown upwards of 6%, as high as 11% even, in recent decades — but it would take quite a lot for an advanced economy like the US to reach 4% from where we are today. Even to reach 3% would take a lot. To grow 3 percent a year for 10 years means that in a decade, the U.S. economy would be a third larger (would be producing a third more) than it is now! Given how much we *already* produce, that's a lot.

Each of the following quotations comes from a real NPR radio show (Planet Money) in which an economist discusses exactly *what* it would take for the US to increase its real GDP growth rate from 2.5% to 4%. For each quotation, indicate **which factor of production** the quotation is referring to. Your choices for factors of production are **labor, capital, human capital, or technology/entrepreneurship**. Choose just **one** factor of production for each quotation.

- (a) “So we would need two dot-com booms going simultaneously, at the same time.” **(3 points)**

- (b) “What if we had... some better education and training?” **(3 points)**

- (c) “We would basically need to double our immigrant population over the next decade.” **(3 points)**

- (d) “We have a lot of signals we send people to tell them to retire early. The Social Security early age is 62. The normal age is 66 and not really going to rise with life expectancy. The Medicare age is 65. You can pull out from your IRA at 59 and a half. We can create tax breaks, like a payroll tax break, for folks that work longer. We could change a lot of the regulations and rules to encourage employers to keep people on for longer.” **(3 points)**

- (e) “...more government building projects, infrastructure, more objects, tools... Hammers, microphones, speakers – I use Microsoft Excel. That’s one of my best pieces of [equipment] in my job.” **(3 points)**

- (f) “What you’d also have to do is take every single adult age 25 to 55, you know, every single prime-age adult and force them to work. And that includes folks that are home because they’re taking care of their kids. That includes folks that are in jail, that are on disability. And it includes some folks that legitimately could be working with the right mix of incentives and support.” **(3 points)**

(g) “If we were to [develop something like the desktop computer] again, if we were – if something were invented that every single office would absolutely have to buy, whether it’s, I don’t know, virtual reality or hologram, something that just transforms business in a way and people have to pour money into it...” **(3 points)**

(h) “We could just all get a whole lot better at our jobs.” **(3 points)**

Problem 2 — Prices from long ago (25 points)

Lupita recently paid a visit to her grandmother’s house. Let’s say Lupita’s grandmother was born in 1927. During her visit, she finds an old shoe box filled with random receipts that her grandmother saved from throughout her life.

The prices on those receipts really astonish her! But you know how to transform them in order to help Lupita think about how those prices relate to the prices that you are familiar with today.

Table 1: The CPI in Different Years

Year	Value of the CPI
1937	14.1
1947	21.5
1950	23.5
1968	34.1
1972	41.1
1980	77.8
1983	100.0
1990	127.4
2005	190.7
2015	233.7
2019	251.7

(a) Which year is the base year according to Table 1? **(1 point)**

- (b) In Lupita's grandmother's shoe box, she finds a receipt for a Reuben sandwich from 1937 when she was just 10 years old! Never mind why she was saving that — the receipt says that the sandwich cost 35 cents. Given inflation in the chart above, how much would the Reuben cost were you to buy it in "today's" 2019 dollars? **(3 points)**
- (c) Lupita also finds a receipt for a gallon of gas from the time of the OPEC crisis of 1980 when people waited in long lines at the pump to buy very expensive gasoline. The receipt says that a gallon of gas cost Lupita's grandmother \$1.25. Given inflation, how much would that same gallon of gas cost were you to buy in "today's" 2019 dollars? **(3 points)**
- (d) Lupita graduated in college from Belmont University in Nashville in 2015. Over four years, her tuition cost her 120,000 in total in 2015 dollars. Lupita's grandmother was the first in her family to graduate from college in 1950 and she saved her tuition receipts. Lupita's grandmother paid 8,000 in 1950 dollars for her four years of education. After accounting for inflation, who paid more for their four years of college? **(3 points)**
- (e) In 1968, Lupita's grandmother bought a very cute Jaguar E Type for \$7,200 for herself on her birthday. Lupita just bought herself a new car in 2015. Hers was a Honda Fit for \$16,200. After accounting for inflation, who paid more for their car? **(3 points)**
- (f) What was the rate of inflation between 1937 and 1947? **(3 points)**
- (g) What was the rate of inflation between 1990 and 2005? **(3 points)**
- (h) **True or False:** The inflation rate that you calculated above between 1990 and 2005 implies that the prices of all goods and services in the economy grew by exactly that amount between those two years? **(1 point)**

- (i) Lupita finds two receipts for Coca Cola in 2005 from two different cities on the same day in her grandmother's shoe box. In St. Louis, the price of a coke was \$1.00. In Chicago, the price of a coke on the same day was \$1.25. If *all else is equal* between the two cities except the price difference between the Cokes, do we say that Chicago has higher inflation than Chicago? Answer **yes or no**, and then briefly **explain your answer**. (5 points)

Problem 3 – Returning to the Production Possibilities Frontier (PPF) model (38 points)

In this problem you will use the production possibilities frontier to think about how economies can grow or not grow. In this problem, we will work with a simplified economy in which there is only one person: Kaitlyn. Kaitlyn can do one of two things with her time. She can either make rabbit traps, which can help Kaitlyn catch rabbits, or she can catch rabbits. Rabbit traps help Kaitlyn catch rabbits, but it's also possible for Kaitlyn to use her two bare hands to catch rabbits as well. (*Think about how you would know that from the plot below if I hadn't told you.*) The particular tradeoff between how she spends her time is given in the table below.

Important notes:

- Assume that if Kaitlyn builds rabbit traps in period 1, those rabbit traps still exist forever into the future, and work just as well. She can use any rabbit traps she produces in period 1 in period 2 and beyond. That is, assume that they do not depreciate over time.
- Assume that nothing else changes between period 1 and 2 *unless the question states otherwise*. That is, hold all else fixed.
- Assume that there are an unlimited number of rabbits in the world. By catching rabbits in period 1, there are not fewer rabbits in period 2.
- Assume that Kaitlyn gets utility from catching and consuming rabbits. The more rabbits she catches, the higher her utility. And she's a rational actor.

Use that information to answer the following questions.

Table 2: Kaitlyn's Production Possibilities Frontier in Period 1

Quantity of Rabbit Traps	Quantity of Rabbits Caught
0	7
1	6
2	4
3	2
4	0

- (a) Which of the two goods that Kaitlyn can produce is a capital good? Which is a consumption good? **(1 point)**
- (b) Draw a production possibilities frontier for Kaitlyn given the information contained in the table above. Put the quantity of **traps on the Y axis** and the quantity of **rabbits on the X axis**. **(5 points)**
- (c) Calculate the slope of Kaitlyn's PPF between 6 and 7 rabbits caught. The slope of a line is the difference in the rise or vertical distance traveled by that line, divided by the distance traveled over the "run" or horizontal direction. The equation for the slope of a line is:

$$slope = \frac{y_2 - y_1}{x_2 - x_1}$$

where y are the y-axis coordinates, and x are the x-axis coordinates. The subscripts refer to which of the two points we are using, point 1 or point 2. It doesn't matter which point we designate as point 1 or point 2. As an example, the slope of the line segment on the PPF between 4 and 6 rabbits caught is the following:

$$slope = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 1}{4 - 6} = \frac{1}{-2} = -\frac{1}{2}$$

Remember that the X axis is the number of rabbits, and the Y axis is the number of rabbit traps. **(5 points)**

- (d) Assume that Kaitlyn only has the hours of sunlight during a day to make rabbit traps or to hunt for rabbits. That is, she's constrained in how much she can produce in any period by the amount of time in her day. (Her time is a scarce resource.) Given the two slopes that you calculated above (or the one you calculated and the one that I calculated as an example), where on the PPF is the marginal opportunity cost of catching an additional rabbit *higher*? That is, where does Kaitlyn have to give up making more rabbit traps to catch an additional rabbit? Briefly **explain** why you think that the difference in opportunity cost exists. You don't need information beyond what is given in the problem set to answer this question. **(5 points)**
- (e) Assume that this is a market economy. In a market economy, Kaitlyn gets to choose how she chooses to allocate her time between producing rabbit traps and catching rabbits. Assume for this part of the problem that she chooses *not* to produce *any* rabbit traps in period 1, and spends all of her resources catching rabbits. What will her PPF look like in **period 2**, all else equal? Put differently, if Kaitlyn uses all of her time to catch rabbits in period 1, will she be able to catch more, less, or the same number of rabbits in period 2 for a given number of traps she produces if everything else remains the same? To illustrate the change, **draw a new PPF**. If the change is *more* or *less* (rather than everything remains the same), there isn't enough information to know *how much more* or *how much less*. So, just draw an *example* of what the PPF *could* look like. *Pay careful attention to the values where the PPF meets each of these axes, however, as you may have enough information in the problem to know what one or both values are.* **(5 points)**
- (f) Assume that once again Kaitlyn decides to use all of her resources in period 1 catching rabbits. Assume also that Kaitlyn learns how to *better* catch rabbits as a result of catching rabbits in period 1. That is, she becomes more skilled at catching rabbits when period 2 rolls around, all else equal. Is this an example of an increase in: **capital, human capital, technology, labor**, or **none of the above**? Choose one answer. **(1 point)**
- (g) Assume as in (f) above that Kaitlyn chooses not to produce any rabbit traps in period 1 and spends all of her time catching rabbits, and that, as in (f), she becomes better at catching rabbits. What will her PPF look like in period 2? Put differently, will she be able to catch more, less, or the same number of rabbits for a given number of rabbit traps she produces in period 2, all else equal? To illustrate the change, draw a new PPF. If the change is more or less (as opposed to everything staying the same), there isn't enough information to know *how much more* or *how much less*. So, just draw an example of what the PPF could look like. *Pay careful attention to the values where the PPF meets each of these axes, however, as you may have enough information in the problem to know what one or both values are.* **(5 points)**

- (h) Assume that we're back in the world where Kaitlyn does NOT become more skilled at catching rabbits between periods. Assume, however, that in period 1 she chooses to produce 2 rabbit traps, in which case she can catch 4 rabbits (according to the PPF). Is this an example of an increase in: **capital**, **human capital**, **technology**, **labor**, or **none of the above**? Choose one answer. (1 point)
- (i) Assume as in (g) above that in period 1 Kaitlyn produces 2 traps and catches 4 rabbits. Can she produce more, less or the same rabbits in period 2 for a given number of rabbit traps produced. Remember that rabbit traps she produces in period 1 still can be used in period 2. To illustrate the change, draw a new PPF. If the change is more or less (as opposed to everything staying the same), there isn't enough information to know *how much more* or *how much less*. So, just draw an example of what the PPF could look like. *Pay careful attention to the values where the PPF meets each of these axes, however, as you may have enough information in the problem to know what one or both values are.* (5 points)
- (j) Assume again that in period 1 Kaitlyn produces 2 rabbit traps, and catches 4 rabbits. All else equal, in period 2 the rabbits become smarter and are less likely to get caught in Kaitlyn's traps than in period 1. In effect, they figure out how to avoid the traps. Is this an example of an increase in: **capital**, **human capital**, **technology**, **labor**? Choose one answer. Notice that "none of the above" is no longer a choice; therefore it *is* one of those four options. This is a bit tricky, but I definitely think that you can figure it out! (2 points)
- (k) Imagine that Kaitlyn lives in Canada, but her doppelganger, Klarissa, who has the exact same PPF as Kaitlyn, lives in Cameroon in Africa. So, Klarissa is exactly the same as Kaitlyn in how many rabbit traps she can produce, how many rabbits she can catch, but she lives in a different country.
- The governments of each of these two countries are the same in every way, except that they have different tax policies. In Canada, the government taxes a small fraction of the rabbits that its citizen catches. It's a small fraction of a rabbit and the tax is always the same.
- In Cameroon, the government is a lot more capricious. Sometimes the government taxes a small fraction of a rabbit, but very frequently it takes all of the rabbits that its citizen produces.
- Consider the decision of Kaitlyn and Klarissa to choose how to allocate their scarce resource (time); that is, how many rabbit traps to produce in period 1 and how many rabbits. Given this information, in which country would you have a greater incentive to invest in the fertilizer? And in which country, therefore, would you expect GDP to be larger? Explain *briefly* why. Note: You may find it helpful to read about "institutions" in the textbook. (5 points)

(1) Something else (**3 points**)

Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.